10. Take a photo from android Camera

Android Studio

Take a photo from android Camera

How to Open Camera Through Intent and Display Captured Image in Android?

Create a New Project in Android Studio

Step 1: Create a xml file for setting file paths

Step 2: Change the manifest file

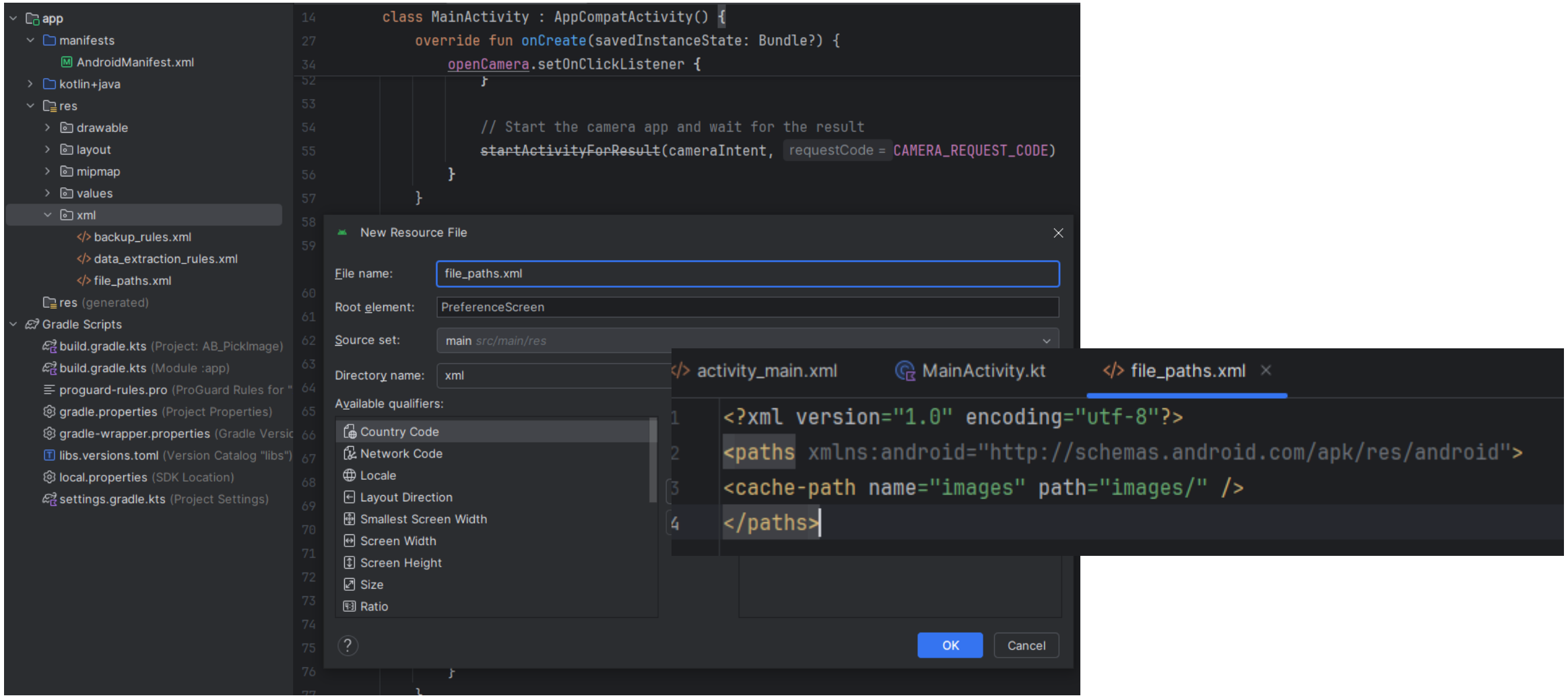
Step 3: Add dependency for image loading

Step 4: Change the activity_main.xml

Step 5: Change the MainActivity File

https://www.geeksforgeeks.org/android/how-to-open-camera-through-intent-and-display-captured-image-in-android/

Step 1: Create a xml file for setting file paths



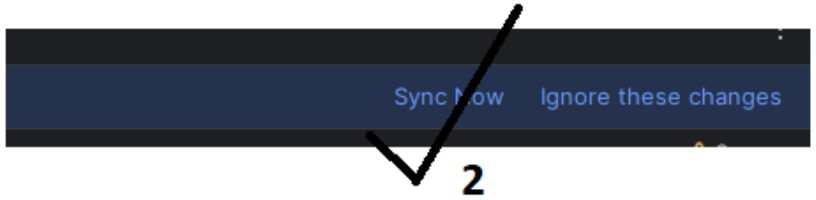
Step 2: Working with manifest file

```
    <activity
        android:name=".MainActivity"
        android:exported="true">
        <intent-filter>
            <action android:name="android.intent.action.MAIN" />

            <category android:name="android.intent.category.LAUNCHER" />
        </intent-filter>
    </activity>

    <provider
        android:name="androidx.core.content.FileProvider"
        android:authorities="${applicationId}.fileprovider"
        android:exported="false"
        android:grantUriPermissions="true">
        <meta-data
            android:name="android.support.FILE_PROVIDER_PATHS"
            android:resource="@xml/file_paths" />
    </provider>
</application>
```

Step 3: Add dependency for image loading




```
dependencies {
    implementation(libs.androidx.core.ktx)
    implementation(libs.androidx.appcompat)
    implementation(libs.material)
    implementation(libs.androidx.activity)
    implementation(libs.androidx.constraintlayout)
    implementation(libs.androidx.material3)
    implementation(libs.androidx.compose.material)
    testImplementation(libs.junit)
    androidTestImplementation(libs.androidx.junit)
    androidTestImplementation(libs.androidx.espresso.core)
    implementation("com.github.bumptech.glide:glide:4.16.0")
}
```

1

Step 4: Working with activity_main.xml

```xml
<!-- Camera Button -->
<Button
    android:id="@+id/camera_open"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Open Camera"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginTop="32dp" />

<!-- ImageView for Captured Image -->
<ImageView
    android:id="@+id/click_image"
    android:layout_width="0dp"
    android:layout_height="0dp"
    app:layout_constraintTop_toBottomOf="@id/camera_open"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintBottom_toBottomOf="parent"
    android:layout_margin="16dp" />
```

Step 5: Working with the MainActivity File

```
1   package com.ab_idekhail.ab_pickimage
2
3   import android.content.Intent
4   import android.net.Uri
5   import android.os.Bundle
6   import android.provider.MediaStore
7   import android.widget.Button
8   import android.widget.ImageView
9   import androidx.appcompat.app.AppCompatActivity
10  import androidx.core.content.FileProvider
11  import com.bumptech.glide.Glide
12  import java.io.File
13
    2 Usages
14  class MainActivity : AppCompatActivity() {
15
16      // Declare UI elements
        2 Usages
17      private lateinit var openCamera: Button
        2 Usages
18      private lateinit var clickedImage: ImageView
19
20      // Variables to hold the file and its Uri
        2 Usages
21      private lateinit var photoFile: File
        3 Usages
22      private lateinit var photoUri: Uri
23
24      // Unique request code for identifying the camera intent result
        2 Usages
25      private val CAMERA_REQUEST_CODE = 1001
```

```
override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)
    setContentView(R.layout.activity_main)

    openCamera = findViewById( id = R.id.camera_open)
    clickedImage = findViewById( id = R.id.click_image)

    openCamera.setOnClickListener {
        // Create a temporary image file to store the captured photo
        photoFile = createImageFile()

        // Get a content URI for the file using FileProvider
        // This allows secure sharing of the file with the camera app
        photoUri = FileProvider.getUriForFile(
            context = this,
            authority = "${packageName}.fileprovider", // FileProvider authority (defined in manifest)
            photoFile
        )

        // Create an intent to launch the camera app
        val cameraIntent = Intent( action = MediaStore.ACTION_IMAGE_CAPTURE).apply {
            // Tell the camera app where to save the image
            putExtra( name = MediaStore.EXTRA_OUTPUT, value = photoUri)
            // Grant temporary write permission to the camera app for the URI
            addFlags( flags = Intent.FLAG_GRANT_WRITE_URI_PERMISSION)
        }

        // Start the camera app and wait for the result
        startActivityForResult(cameraIntent, requestCode = CAMERA_REQUEST_CODE)
    }
}
```

```
// Function to create a temporary image file in the cache directory
1 Usage
private fun createImageFile(): File {
    // Create a directory inside cache to store images (if not already created)
    val imageDir = File( parent = cacheDir,  child = "images").apply { mkdirs() }
    // Create a temp file with prefix "captured_" and suffix ".jpg"
    return File.createTempFile( prefix = "captured_",  suffix = ".jpg",  directory = imageDir)
}

// Deprecated but still usable for handling activity results (like camera)
@Deprecated( message = "Deprecated in Java")
override fun onActivityResult(requestCode: Int, resultCode: Int, data: Intent?) {
    super.onActivityResult(requestCode, resultCode, data)

    // Check if the result is from the camera and was successful
    if (requestCode == CAMERA_REQUEST_CODE && resultCode == RESULT_OK) {
        // Load the captured image from the URI into the ImageView using Glide
        Glide.with( activity = this).load(photoUri).into( view = clickedImage)
    }
}
```

HW3: How to Capture Screenshot of a View and Save it to Gallery in Android?

https://www.geeksforgeeks.org/android/how-to-capture-screenshot-of-a-view-and-save-it-to-gallery-in-android/